Challenge: Towards UWB-Enabled Spatial Healthcare

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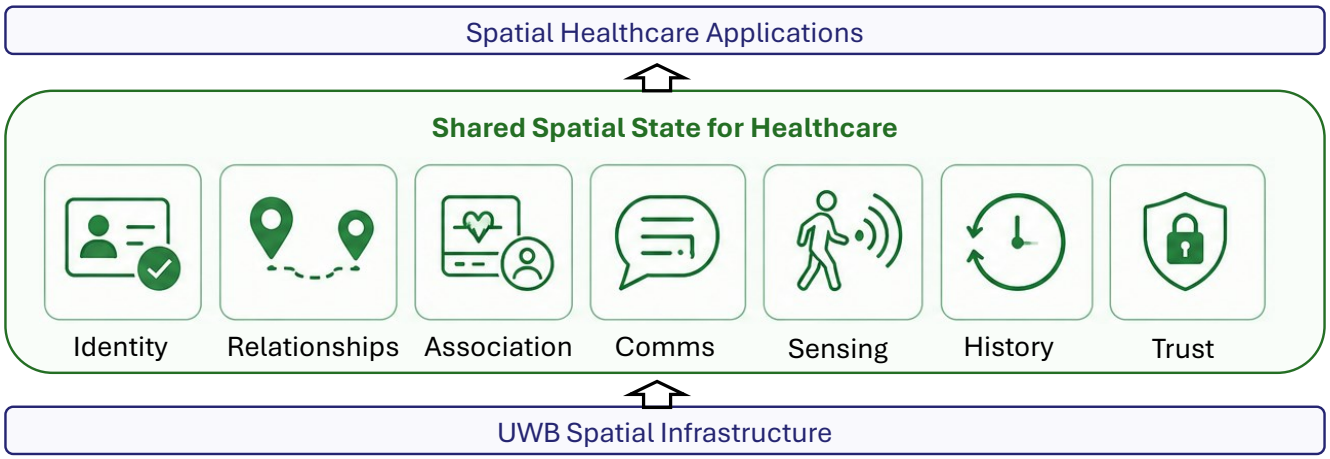


Figure 1: Our UWB-Enabled Spatial Healthcare Vision

Abstract

Healthcare workflows depend on evolving relationships among patients, caregivers, devices, and clinical spaces, yet existing systems typically treat location as application-specific metadata rather than shared infrastructure. We envision *spatial context* as a first-class systems primitive that supports identity, relationships, association, communication, sensing, history, and trust among healthcare entities. Ultra-wideband (UWB) is a promising foundational technology for this vision as it enables fine-grained ranging, communication, secure distance-measurement mechanisms, and emerging sensing capabilities on a common radio platform. This paper outlines the spatial services required for a UWB-enabled spatial healthcare platform, the novel applications it might enable, and the multidisciplinary challenges that must be addressed to realize this vision.

1 Introduction

Modern healthcare environments are shaped by relationships among patients, caregivers, medical devices, time, and hospital infrastructure. Measurements must be associated with the correct patient; mobile equipment must be available where care is delivered; infection-control investigations depend on reconstructing meaningful contacts; and mobility itself can be an important indicator of rehabilitation and recovery [1, 11, 12]. As a result, healthcare environments increasingly deploy real-time locating systems for individual tasks such as asset tracking, patient flow, and contact tracing [5]. However, these systems commonly expose application-specific location coordinates rather than a shared representation of the evolving spatiotemporal relationships between people and medical devices.

We envision spatial context as a first-class systems primitive for healthcare. Instead of requiring each application to independently reconstruct location and proximity, a common spatial infrastructure would maintain identities, relationships, associations, and histories and expose them through reusable programming abstractions. This vision extends beyond conventional localization; the objective is not merely to estimate coordinates but to determine, for example, whether a nearby monitor is currently associated with a particular patient, whether a contact was clinically meaningful, and whether the available spatial information is sufficiently reliable to support an action or inference. Prior work on augmenting electronic health records with real-time location illustrates both the value of spatial information and the absence of such a common substrate [8, 12].

Healthcare presents an unusually challenging setting for this infrastructure. Spatial relationships are dynamic and temporary; participating devices are heterogeneous; incorrect associations can affect patient safety; and spatial histories reveal sensitive information about patients and healthcare workers. Furthermore, hospitals have a high density of walls, people, metal equipment, and changing layouts that create non-line-of-sight propagation, multipath, and body blockage

[4, 13, 14]. A useful system must therefore combine fine-grained spatial measurements with identity, clinical semantics, environmental boundaries, uncertainty, security, and privacy. *These challenges distinguish spatial healthcare from a typical indoor-localization problem.*

Considering the mentioned challenges, Ultra-wideband (UWB) technology can be a promising foundation for this vision, and its growing standardization and commercial availability make such an infrastructure increasingly practical. Its time-of-flight measurements provide fine-grained ranging, the IEEE 802.15.4z standard adds mechanisms intended to strengthen ranging integrity [7], and commercial radios can expose channel impulse responses that support fine-grained wireless sensing [2, 9]. While other wireless technologies, e.g., Wi-Fi and Bluetooth, can provide subsets of UWB capabilities, UWB's appeal lies in the possibility of deriving communication, ranging, localization, and sensing from a shared radio platform. This convergence is key: capabilities that previously required separate, application-specific systems can now be explored as components of a common spatial infrastructure. Although UWB does not, by itself, establish clinical identity, infer meaningful associations, or guarantee trustworthy operation for spatial healthcare, it provides a particularly useful substrate on which higher-level spatial services can be built to realize our vision.

This paper presents a vision for a UWB-enabled spatial infrastructure for healthcare, illustrated in Fig. 1. In the rest of the paper, we explain why healthcare requires spatial context rather than isolated location estimates. We then identify several spatial services that such an infrastructure should expose to maintain a full spatial context. Finally, we outline some of the multidisciplinary research challenges that must be addressed to realize the vision.

2 Why Healthcare Needs Spatial Context

In this section, we show how healthcare is organized around spatial relationships, motivating the need for spatial context. First, healthcare is delivered to a particular patient by certain caregivers and instruments, at particular **places**, often while the patient **moves** across rooms, departments, facilities, and their home. Thus, a physiological measurement is useful only when it is associated with the correct **patient, device, and time**. In addition, infection-control investigations depend on reconstructing meaningful co-presence among patients and healthcare workers [1]. Moreover, mobile healthcare equipment and medications must be available *where they are needed*. Furthermore, rehabilitation and recovery are often evaluated through movement and mobility; walking speed, for example, is widely used as an indicator of functional status and overall health [11]. Thus, many healthcare activities depend not only on the locations of individual entities but also on the *evolving spatial relationships* among people, devices, places, and clinical events.

As a result, today's healthcare systems try to maintain location information, with electronic health records (EHRs) storing room, bed, department, and encounter locations, enabled by hospitals increasingly deploying real-time localization systems [8, 15]. Nevertheless, an architectural limitation remains: location is commonly represented as metadata, e.g., a field within an application, or supplied by a task-specific tracking system, rather than maintained as a shared, continuously updated organizing principle. The need to augment an EHR with real-time location services [8], and separate efforts to automate patient-device association and disassociation [12], illustrate this gap. Similarly, existing localization systems research has largely evaluated applications such as asset tracking, patient flow, workflow analysis, and contact tracing **independently** [1], lacking a common substrate that can relate identities, devices, measurements, places, and interactions as they evolve.

We thus argue that spatial context in healthcare should become a first-class systems primitive rather than metadata reconstructed independently by each application. Such an infrastructure should allow direct reasoning about questions such as which device is currently associated with a patient, which caregiver is responsible for an observation, or which contacts were clinically meaningful during a given interval. *Communication, ranging, localization, sensing, and identity then become complementary mechanisms for constructing and maintaining this shared spatial state.* The next section identifies the services that such an infrastructure should expose.

3 UWB-Enabled Spatial Healthcare Services

The goal of this infrastructure is to expose spatial context through a common set of reusable services. UWB is a compelling substrate because it combines communication, ranging, localization, and sensing on a shared radio platform. The remainder of this section describes the services built upon this foundation.

Spatial Identity. Traditional digital identity establishes *who* is participating in a system through identifiers such as device addresses. A spatial healthcare infrastructure must additionally establish *where* those physical entities are located. Spatial identity should therefore maintain a continuously updated binding between digital identities and physical entities. Unlike conventional localization, which typically returns only a coordinate, this service enables applications to infer that a credentialed device corresponds to a particular patient, caregiver, or medical asset as it moves through the environment. Because identity errors can affect patient safety, proximity must supplement rather than replace clinical identification.

UWB couples authenticated device credentials with fine-grained distance evidence through secure ranging. While

clinical identity still requires enrollment and authentication, UWB strengthens the binding between a digital identity and its physical presence [7].

Spatial Relationships. Spatial relationships describe how identified entities are situated relative to one another. Examples include *near*, *inside the same room*, *approaching*, *moving together*, or *separated by a clinical boundary*. Rather than exposing only coordinates, the system should maintain a dynamic graph of typed relationships and attach timestamps and confidence values, instead of requiring every application to reconstruct them independently from coordinates.

UWB provides precise time-of-flight distance measurements, offering strong evidence for spatial relationships. However, clinically meaningful relationships cannot be inferred from distance alone; floor plans, room boundaries, temporal reasoning, and application semantics are required to distinguish true interactions from simple proximity [6].

Spatial Association. Healthcare continuously produces measurements, observations, and interactions that must be associated with the correct patient, caregiver, device, or location. Given spatial identities and relationships, a spatial healthcare infrastructure should automatically decide which entities, measurements, and clinical events belong together. Associations must be created, updated, and revoked while handling entity mobility and ambiguous evidence.

UWB provides accurate ranging that can trigger association and disassociation as entities move. However, deciding whether measurements should be associated with a patient also requires identity, device capabilities, workflow state, temporal stability, and clinical policies. These decisions therefore belong above the radio and MAC layers, which need further innovations.

Spatial Communication. Spatial communication allows applications to address a spatial role or neighborhood rather than a fixed network endpoint. Operations such as `send_to(patient_hub)` or `notify(nearest_authorized_caregiver)` could use the current spatial state to select recipients, enforce access controls, facilitate discovery, tune transmission power, and allocate resources. This is particularly useful in large hospitals where many mobile devices coexist, and communication permissions depend on clinical context.

Because UWB combines ranging and low-rate communication on the same radio, it naturally supports spatially scoped communication among nearby devices. Exploiting this capability for interference reduction and spatial reuse, however, requires spatially-aware scheduling, power control, and coexistence mechanisms [7].

Spatial Sensing. Spatial sensing infers properties of the environment without requiring every observed person or object to carry a tag. In healthcare, it could expose higher-level observations such as presence, mobility, and environmental changes, while hiding raw radio measurements from applications. Such observations must include an uncertainty measure and be governed by appropriate privacy and access-control policies.

Commercial UWB radios expose channel impulse response (CIR) measurements that support fine-grained device-free sensing. Combined with UWB's large bandwidth, these measurements can detect subtle changes in multipath and human motion, providing a common sensing substrate alongside communication and ranging.

Spatial History. Healthcare interactions unfold over time, making temporal history an essential component of spatial context. A spatial healthcare infrastructure should maintain histories of how identities, relationships, and associations evolve over time, supporting contact investigations, clinical workflow analysis, mobility assessment, safety auditing, and retrospective studies. They must represent intervals, uncertainty, and changes.

UWB provides timestamped range and CIR observations that can be used to construct spatial histories without continuously storing coordinates. CIR measurements may also help distinguish meaningful same-room interactions from devices that are physically close but separated by walls [3].

Spatial Trust. Spatial trust communicates how much confidence an application should place in spatial information. It should expose the provenance, uncertainty, authenticity, and consistency of ranges, relationships, associations, and sensed events. A safety-critical application may then demand stronger evidence than an application performing equipment inventory or workflow analysis.

IEEE 802.15.4z strengthens UWB ranging integrity through secure ranging mechanisms that resist distance-manipulation attacks [7]. Combined with authenticated identities, redundant observations, and application-level safety policies, these mechanisms provide a strong foundation for trustworthy spatial interactions in healthcare.

4 Multidisciplinary Challenges

Realizing the services described above requires addressing multidisciplinary challenges spanning wireless systems, distributed systems, security, and clinical practice.

Reliable spatial inference in clinical environments. Hospitals are particularly challenging environments, with a high density of walls, mobile equipment, people, and changing room configurations creating non-line-of-sight (NLOS) and multipath conditions. Although UWB is relatively resilient in these conditions, these effects can still bias its range estimates [4, 13, 14]. Systems must therefore detect degraded links, fuse observations across anchors, and expose uncertainty rather than only returning a coordinate. More importantly, systems must reliably distinguish clinically meaningful boundaries such as adjacent beds, rooms,

and patient-device associations. Achieving this reliability while supporting many mobile devices, practical update rates, and battery-powered wearables remains an open challenge.

Spatial representation and semantics. Coordinates alone cannot capture the healthcare meaning of space. A healthcare spatial infrastructure must therefore represent typed relationships such as *inside-room*, *near-patient*, *assigned-to*, and *observed-by*, together with their temporal context, provenance, and uncertainty. It must also distinguish physical proximity from clinical association. The challenge is thus to translate noisy range measurements into meaningful, efficient, and queryable relationships.

Programming abstractions and distributed state. Today's communication abstractions are fundamentally device-centric. Spatial healthcare should reason about entities and relationships rather than only radio identifiers or coordinates. For example, instead of `send(IP, Port)`, an application might require operations in which the destination is dynamically determined through spatial association, like `send_to(associated_patient_hub)` or `search_nearby(type_ultrasound)`. Implementing such abstractions is difficult because the spatial state is distributed, where devices observe only partial views that may be delayed, noisy, or contradictory. The infrastructure must determine where the state is maintained, how observations are reconciled, and what history should persist. Designing abstractions that hide this complexity without hiding clinically important uncertainty is a central research challenge.

Security, privacy, and safety. Secure UWB ranging strengthens evidence that two authenticated radios are at a particular distance, but it does not establish that a radio is attached to the correct patient, authorize the use of its location, or guarantee that the resulting clinical association is correct. Moreover, practical attacks demonstrate that cryptographically protected UWB ranging is insufficient on its own [7, 10]. Spatial healthcare, therefore, requires credential binding, compromised-device detection, redundant observations, and fail-safe behavior. In addition, spatial histories can reveal sensitive routines, interactions, and health-related behavior. Healthcare environments require strong privacy guarantees that prevent unnecessary disclosure of patient data and behavioral patterns while still enabling clinically useful spatial reasoning. Balancing trustworthy spatial information with privacy preservation remains a fundamental challenge.

Deployment and clinical adoption. A spatial infrastructure must integrate with existing medical devices, electronic health records, and clinical workflows while imposing minimal burden on patients and staff. Practical deployments must address anchor installation and calibration, battery maintenance, tag attachment, false alerts, and changes to hospital layouts. Moreover, new systems must demonstrate measurable improvements in patient outcomes, workflow efficiency, and operational cost while satisfying regulatory requirements and maintaining compatibility with existing healthcare information systems [5]. Close collaboration among systems researchers, clinicians, administrators, and patients is essential for translating spatial healthcare from infrastructure prototypes into routine practice.

5 Conclusion

Rather than viewing localization as a standalone application, we argue that it should be a reusable system service that supports higher-level spatial reasoning across healthcare. UWB is particularly compelling because it offers a common radio platform from which many of these services can be constructed. We hope this vision encourages future research toward programmable spatial healthcare infrastructure, where applications no longer reconstruct spatial context independently but instead build upon a shared spatial substrate.

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